

A crystallographic restriction for multiplier cyclic sieving on necklaces

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Abstract

Let $Y_{n,k}(q) = [n]_q / [k]_q$ be the natural q -analogue of the number $\binom{n}{k} / n$ of binary necklaces of length n with k black beads, defined whenever $\gcd(n, k) = 1$. Rotation acts trivially on necklaces, but the *multipliers* $x \mapsto vx$ for $v \in (\mathbb{Z}/n\mathbb{Z})^\times$ do not; group-theoretically these are the symmetries lying in the normaliser of the rotation group inside S_n . We ask when $Y_{n,k}$ evaluated at a root of unity counts the necklaces a multiplier fixes.

Writing $f_e = \gcd(v^e - 1, n)$ for the number of points of $\mathbb{Z}/n\mathbb{Z}$ held fixed by v^e , the criterion reads: the sieving identity holds for every k coprime to n precisely when, for every proper divisor e of the order m of v , one has $\varphi(f_e) \leq 2$ — that is $f_e \in \{1, 2, 3, 4, 6\}$ — together with $\text{rad}(f_e) \mid m$. We prove that these conditions are *sufficient*, for every multiplier and with no uniformity hypothesis (Theorem 6.3).

Three ingredients make this possible. First, an unconditional collapse: every v -fixed necklace carries exactly f_1 representatives fixed by v as words, so that counting fixed necklaces reduces to the cycle index of a single permutation (Theorem 3.2); this holds for every colour content, not only for two colours. Second, a structure theorem: under the criterion the points of $\mathbb{Z}/n\mathbb{Z}$ whose v -orbit is shorter than m form a cyclic subgroup C_L with $L \in \{1, 2, 3, 4, 6\}$ on which v acts as ± 1 (Theorem 7.2). Third, the observation that the orbit polynomial $\prod_{|O| < m} (1 + z^{|O|})$ is palindromic of degree L with linear coefficient f_1 , while the residues $k \bmod m$ attainable subject to $\gcd(n, k) = 1$ are exactly $\{1, L - 1\}$ — a statement equivalent to $\varphi(L) \leq 2$. The crystallographic restriction is thus not an input to the argument but a reformulation of its hypothesis.

We also correct the record. The irrationality hypothesis on which the converse direction was previously made to rest is false: the smallest counterexample is $m = 9$, $r = 8$, $s = 4$, where the relevant ratio equals -1 , and there are 67 with $m \leq 60$, all at $r = m - 1$ and all of modulus 1. We prove the congruence $m \mid r - 1 - s(r - s)$ that rationality forces, evaluate the ratio in closed form at $r = m - 1$, and repair the necessity argument accordingly. The first multiplier at which the gap is felt is $n = 99$, $v = 19$ — just beyond the range $n \leq 96$ over which the criterion had been checked — where $Y_{99,13}(\zeta_{10}) = -9$ while the true count is 84. All computations are exact. The cases $f \in \{3, 4, 6\}$ lie in a regime that Stucky [2] explicitly leaves open.

1 Introduction

Looping a string makes a cyclic group act on it, and the characters of a cyclic group are roots of unity. The sharpest modern expression of that principle is the *cyclic sieving phenomenon* of Reiner, Stanton and White [1]; see [3] for a survey.

*Discovered, verified and drafted in collaboration with Claude (Anthropic). All computations are exact and reproducible from the accompanying source code; see Section 10.

Definition 1.1 ([1]). Let X be a finite set carrying an action of a cyclic group $C = \langle c \rangle$ of order N , and let $X(q) \in \mathbb{Z}[q]$ satisfy $X(1) = |X|$. The triple $(X, X(q), C)$ exhibits the *cyclic sieving phenomenon* if for every $d \in \mathbb{Z}$,

$$|\{x \in X : c^d x = x\}| = X(q)|_{q=\zeta_N^d},$$

where $\zeta_N = e^{2\pi i/N}$.

Fix $n \geq 3$ and colour the positions $\mathbb{Z}/n\mathbb{Z}$ with two colours, k of them black. Modulo the rotation action of $\mathbb{Z}/n\mathbb{Z}$ these are the *necklaces*; when $\gcd(n, k) = 1$ the rotation action is free and there are exactly $\binom{n}{k}/n$ of them. The natural q -analogue of that count is

$$Y_{n,k}(q) = \frac{\begin{bmatrix} n \\ k \end{bmatrix}_q}{[n]_q}, \quad [n]_q = 1 + q + \cdots + q^{n-1}, \quad (1)$$

which is a polynomial with non-negative integer coefficients precisely when $\gcd(n, k) = 1$ (Proposition 4.1).

Rotation itself acts trivially on necklaces, so it carries no information. What survives is the action of the *multiplier group* $(\mathbb{Z}/n\mathbb{Z})^\times$: for a unit v , relabel positions by $x \mapsto vx$. Inside S_n these are exactly the permutations normalising the cyclic rotation group, so the full symmetry available is the affine group $\mathbb{Z}/n\mathbb{Z} \rtimes (\mathbb{Z}/n\mathbb{Z})^\times$. Rotation spins a necklace; a multiplier rewires it.

Known results. For n prime the sieving always holds; this is Corollary 3.3 of Stucky [2], whose general theorem covers multipliers acting on $\mathbb{Z}/n\mathbb{Z}$ with exactly one or two fixed points, subject to conditions on the associated Young subgroup. Beyond that the paper stops, and says so:

“... when $n \not\equiv 1, 2 \pmod m$ the situation appears much more delicate, and we do not have a general conjecture.” [2, §3]

That regime — composite n , multipliers with three or more fixed points — is the subject of this note.

Results. Section 3 proves the collapse theorem: the number of multiplier-fixed necklaces is the number of multiplier-fixed *words* divided by f_1 (Theorem 3.2). It holds for every colour content and under no hypotheses, and it is what makes the question decidable in $O(n^2)$ arithmetic operations rather than by exponential enumeration. Section 4 records the cyclotomic factorisation of $Y_{n,k}$ and its evaluation at roots of unity, in both the case $m \nmid n$ and the degenerate case $m \mid n$. Section 5 settles which of these evaluations are rational; the hypothesis previously assumed here is disproved. Section 6 states the criterion, Section 7 reformulates it as a statement about a subgroup, and Section 8 proves sufficiency. Section 9 treats the converse: the fixed-point count is forced into the crystallographic set $\{1, 2, 3, 4, 6\}$. Section 10 reports the computational evidence and Section 11 the boundaries. We are explicit throughout about which statements are proved and which are only verified.

2 Notation

Throughout $n \geq 3$, and $\alpha = (\alpha_1, \dots, \alpha_\ell)$ is a *content*: a composition of n recording how many positions receive each of ℓ colours. Words of content α are maps $w : \mathbb{Z}/n\mathbb{Z} \rightarrow$

$\{1, \dots, \ell\}$ with $|w^{-1}(i)| = \alpha_i$. The rotation group $\mathbb{Z}/n\mathbb{Z}$ acts on them, freely if and only if $\gcd(\alpha) = 1$; the orbits are α -necklaces. The binary case is $\alpha = (k, n - k)$ with $\gcd(n, k) = 1$.

For $v \in (\mathbb{Z}/n\mathbb{Z})^\times$ we write $m = \text{ord}_n(v)$ and let v act on necklaces by $[w] \mapsto [w \circ v]$. We put

$$f_e = \gcd(v^e - 1, n) = |\{x \in \mathbb{Z}/n\mathbb{Z} : v^e x = x\}|, \quad f = f_1.$$

Φ_d denotes the d -th cyclotomic polynomial, ζ_m a primitive m -th root of unity, φ Euler's totient, and $\text{rad}(x) = \prod_{p|x} p$ the radical. We call v *uniform* if every $\langle v \rangle$ -orbit on $\mathbb{Z}/n\mathbb{Z}$ has size 1 or m .

3 Counting multiplier-fixed necklaces

For $c \in \mathbb{Z}/n\mathbb{Z}$ let $A_c : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ be the affine map $A_c(y) = vy + c$; these are exactly the affine permutations with linear part v . Set

$$S_j = 1 + v + v^2 + \dots + v^{j-1} \in \mathbb{Z}/n\mathbb{Z},$$

let $N_\alpha(A_c)$ denote the number of words of content α constant on the cycles of A_c , and abbreviate $N_\alpha(v) = N_\alpha(A_0)$: the number of words of content α constant on the $\langle v \rangle$ -orbits of $\mathbb{Z}/n\mathbb{Z}$, equivalently the number of words w with $w \circ v = w$.

Lemma 3.1. *Let $\gcd(\alpha) = 1$. A necklace $[w]$ of content α is fixed by the multiplier v if and only if w is constant on the cycles of A_c for some $c \in \mathbb{Z}/n\mathbb{Z}$, and that c is then unique.*

Proof. Existence. $[w]$ is fixed by v iff $w \circ v = w \circ \rho_t$ for some rotation $\rho_t(x) = x + t$. Substituting $y = x + t$ turns this into $w \circ A = w$ for the affine map $A(y) = v(y - t)$, whose linear part is v . Conversely any w with $w \circ A_c = w$ has v -fixed necklace class.

Uniqueness. If w is constant on the cycles of both A_c and $A_{c'}$ with $c \neq c'$, then w is invariant under $A_c A_{c'}^{-1}$, which is the rotation by $c - c' \neq 0$. Since $\gcd(\alpha) = 1$ the rotation action on words of content α is free, a contradiction. $\square$

The next theorem is the engine of the paper. It says that only the indices $c \in (v - 1)\mathbb{Z}/n\mathbb{Z}$ occur in Lemma 3.1, and that the resulting correspondence between fixed words and fixed necklaces is uniformly f_1 -to-one. Counting fixed necklaces therefore reduces to the cycle index of the single permutation $y \mapsto vy$. No hypothesis on v , on n or on the number of colours is required.

Theorem 3.2 (collapse). *Let α be any content with $\gcd(\alpha) = 1$ and let $v \in (\mathbb{Z}/n\mathbb{Z})^\times$. Then:*

- (i) $N_\alpha(A_c) = 0$ unless $f_1 \mid c$, that is, unless $c \in (v - 1)\mathbb{Z}/n\mathbb{Z}$;
- (ii) the map $w \mapsto [w]$ from $\{w : w \circ v = w\}$ to the set of v -fixed α -necklaces is surjective, with every fibre of size exactly f_1 ;
- (iii) $\#\text{Fix}(v) = \frac{N_\alpha(v)}{f_1}$.

Proof. (i) Let ℓ be a cycle length of A_c , say through y . Then $A_c^\ell(y) = v^\ell y + cS_\ell = y$, so the congruence $(v^\ell - 1)y \equiv -cS_\ell$ is solvable and hence $f_\ell = \gcd(v^\ell - 1, n)$ divides cS_ℓ .

Since $v - 1$ divides $v^\ell - 1$ we have $f_1 \mid f_\ell$, so $f_1 \mid cS_\ell$; and $v \equiv 1 \pmod{f_1}$ gives $S_\ell \equiv \ell \pmod{f_1}$. Therefore

$$f_1 \mid c\ell \quad \text{for every cycle length } \ell \text{ of } A_c, \quad (2)$$

and consequently $e_c := f_1 / \gcd(f_1, c)$ divides *every* cycle length of A_c . Suppose $f_1 \nmid c$, so that $e_c > 1$, and choose a prime $p \mid e_c$. Then $p \mid f_1 \mid n$, and p divides every cycle length of A_c . A word constant on the cycles of A_c has each colour class a union of cycles, so $p \mid \alpha_i$ for every i and hence $p \mid \gcd(\alpha) = 1$, which is absurd. Thus $N_\alpha(A_c) = 0$. Finally $\{c : f_1 \mid c\} = f_1\mathbb{Z}/n\mathbb{Z}$ is the image of multiplication by $v - 1$, since $\gcd(v - 1, n) = f_1$.

(ii) *Surjectivity*. Let $[w]$ be v -fixed. By Lemma 3.1 there is a c with $w \circ A_c = w$, and by (i) we may write $c = (v - 1)d$. Put $w' = w \circ \rho_{-d}$. Then $w'(vx) = w(vx - d)$ and $w'(x) = w(x - d)$, and substituting $y = x - d$ shows that $w' \circ v = w'$ is equivalent to $w(vy + (v - 1)d) = w(y)$, i.e. to $w \circ A_c = w$. So $w' \in [w]$ is a v -fixed word.

Fibres. Let $w \circ v = w$ and $t \in \mathbb{Z}/n\mathbb{Z}$. The same substitution gives $(w \circ \rho_t) \circ v = w \circ \rho_t$ if and only if $w \circ A_{t(1-v)} = w$, which by the uniqueness clause of Lemma 3.1 forces $t(1 - v) = 0$, i.e. $t \in \text{Fix}(v)$, a set of size f_1 . As the rotation action is free, these f_1 rotates are distinct words, and they are precisely the v -fixed words in $[w]$.

(iii) Immediate from (ii), since $|\{w : w \circ v = w\}| = N_\alpha(v)$. $\square$

Corollary 3.3. *Let $\gcd(n, k) = 1$ and let n_d denote the number of $\langle v \rangle$ -orbits on $\mathbb{Z}/n\mathbb{Z}$ of size exactly d . Then*

$$\# \text{Fix}(v) = \frac{1}{f_1} [z^k] \prod_{d \mid m} (1 + z^d)^{n_d}. \quad (3)$$

Proof. A word of content $(k, n - k)$ constant on the $\langle v \rangle$ -orbits is exactly a choice of a sub-multiset of orbits of total size k , and the generating function for the orbit sizes is $\prod_{d \mid m} (1 + z^d)^{n_d}$. Apply Theorem 3.2(iii). $\square$

Remark 3.4. Theorem 3.2 is strictly stronger than the observation that only c with $cS_m = 0$ can contribute: $(v - 1)\mathbb{Z}/n\mathbb{Z}$ is contained in $\{c : cS_m = 0\}$, usually properly, and the theorem also identifies the common value of the surviving terms. Computationally (3) replaces a sum of cycle-index computations by a single $O(n)$ orbit decomposition followed by one coefficient extraction.

4 The q -analogue at a root of unity

Proposition 4.1. *Let $\gcd(n, k) = 1$. Then $Y_{n,k}(q)$ of (1) is a polynomial, and*

$$Y_{n,k}(q) = \prod_{d \in E(n,k)} \Phi_d(q), \quad E(n,k) = \{d \geq 2 : d \nmid n \text{ and } (k \bmod d) > (n \bmod d)\}.$$

Proof. From $[j]_q = \prod_{d \mid j, d \geq 2} \Phi_d(q)$ we get $[n]_q! = \prod_{d \geq 2} \Phi_d^{\lfloor n/d \rfloor}$, hence $\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{d \geq 2} \Phi_d^{\varepsilon_d}$ with $\varepsilon_d = \lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n - k)/d \rfloor$. Writing $a = k \bmod d$, $b = (n - k) \bmod d$, $c = n \bmod d$, we have $a + b \in \{c, c + d\}$ and $\varepsilon_d = (a + b - c)/d \in \{0, 1\}$, with $\varepsilon_d = 1$ iff $b = c - a + d$, i.e. iff $a > c$. Also $[n]_q = \prod_{d \mid n, d \geq 2} \Phi_d$. For $d \mid n$ with $d \geq 2$ we have $c = 0$ and, since $\gcd(n, k) = 1$ forces $d \nmid k$, also $a > 0 = c$; so $\varepsilon_d = 1$ and every factor of $[n]_q$ is cancelled exactly. The quotient is the stated product. $\square$

Proposition 4.2 (q -Lucas). *Let ζ be a primitive m -th root of unity, and write $n = am + r$, $k = bm + s$ with $0 \leq r, s < m$. Then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=\zeta} = \begin{pmatrix} a \\ b \end{pmatrix} \begin{bmatrix} r \\ s \end{bmatrix}_q \Big|_{q=\zeta}, \quad [n]_q \Big|_{q=\zeta} = [r]_\zeta.$$

Consequently, if $m \nmid n$,

$$Y_{n,k}(\zeta) = \begin{pmatrix} a \\ b \end{pmatrix} \cdot \frac{[s]_q \Big|_{q=\zeta}}{[r]_\zeta}. \quad (4)$$

Proof. The first identity is the q -Lucas theorem. The second holds because $\sum_{i=0}^{n-1} \zeta^i$ splits into a complete cycles, each summing to 0 when $m > 1$, leaving $\sum_{i=0}^{r-1} \zeta^i$. $\square$

Corollary 4.3. *With $r \geq 1$ and $0 \leq s$, the ratio in (4) satisfies*

$$\frac{[s]_q \Big|_{q=\zeta}}{[r]_\zeta} = \begin{cases} 1, & s \in \{1, r-1\}, \\ 0, & s > r, \\ 1/[r]_\zeta, & s \in \{0, r\}. \end{cases}$$

Proof. $\begin{bmatrix} r \\ 1 \end{bmatrix}_q = \begin{bmatrix} r \\ r-1 \end{bmatrix}_q = [r]_q$; $\begin{bmatrix} r \\ 0 \end{bmatrix}_q = \begin{bmatrix} r \\ r \end{bmatrix}_q = 1$; and $\begin{bmatrix} r \\ s \end{bmatrix}_q = 0$ for $s > r$. $\square$

Thus (4) is manifestly a rational integer for $s \in \{1, r-1\}$ and vanishes for $s > r$. We call the remaining residues

$$B(r) = \{0, 2, 3, \dots, r-2, r\}$$

the *bad* residues. For these the ratio is an algebraic number in $\mathbb{Q}(\zeta)$ whose rationality is a delicate matter, taken up in Section 5.

Proposition 4.2 degenerates when $m \mid n$, for then $r = 0$ and $[n]_\zeta = 0$. That case is governed by the following variant, which will be needed in Section 8.

Proposition 4.4. *Suppose $m \mid n$ and $\gcd(n, k) = 1$, and set $a = n/m$, $b = \lfloor k/m \rfloor$, $s = k \bmod m$. Then $s \geq 1$ and*

$$Y_{n,k}(\zeta_m) = \begin{pmatrix} a-1 \\ b \end{pmatrix} \sigma_s, \quad \sigma_s = \frac{(-1)^{s-1} \zeta^{-s(s-1)/2}}{[s]_\zeta}. \quad (5)$$

Proof. From $\begin{bmatrix} n \\ k \end{bmatrix}_q = ([n]_q/[k]_q) \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$ we obtain the polynomial identity $Y_{n,k}(q) = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q/[k]_q$, which avoids the vanishing of $[n]_q$ at ζ . Since $m \mid n$ and $\gcd(n, k) = 1$ we have $\gcd(k, m) = 1$, so $s \geq 1$ and $[k]_\zeta = (1 - \zeta^k)/(1 - \zeta) = (1 - \zeta^s)/(1 - \zeta) = [s]_\zeta \neq 0$. Writing $n-1 = m(a-1) + (m-1)$ and $k-1 = mb + (s-1)$, Proposition 4.2 gives $\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q \Big|_\zeta = \begin{pmatrix} a-1 \\ b \end{pmatrix} \begin{bmatrix} m-1 \\ s-1 \end{bmatrix}_q \Big|_\zeta$. Finally, for $0 \leq j \leq m-1$,

$$\begin{bmatrix} m-1 \\ j \end{bmatrix}_q \Big|_{q=\zeta} = \prod_{i=1}^j \frac{1 - \zeta^{m-1-j+i}}{1 - \zeta^i} = \prod_{u=1}^j \frac{1 - \zeta^{-u}}{1 - \zeta^u} = (-1)^j \zeta^{-j(j+1)/2}, \quad (6)$$

the middle equality by the substitution $u = j+1-i$ (legitimate since $m-1-j+i \equiv -(j+1-i) \bmod m$) and the last by $1 - \zeta^{-u} = -\zeta^{-u}(1 - \zeta^u)$. Take $j = s-1$. $\square$

5 Rationality of the ratio

Fix a primitive m -th root of unity ζ and integers $0 \leq s \leq r < m$, and put

$$\rho_{r,s} = \frac{[s]_q|_{q=\zeta}}{[r]_\zeta} \in \mathbb{Q}(\zeta).$$

This is defined and non-zero: $[r]_\zeta \neq 0$ since $m \nmid r$ for $1 \leq r < m$, and $[s]_q|_\zeta \neq 0$ since, by the computation in the proof of Proposition 4.1, the factor Φ_m divides $[s]_q$ only when $s \bmod m > r \bmod m$, that is only when $s > r$.

Proposition 5.1 (conjugation constraint). *If $\rho_{r,s} \in \mathbb{Q}$ then $m \mid r - 1 - s(r - s)$.*

Proof. Complex conjugation on $\mathbb{Q}(\zeta)$ sends $\zeta \mapsto \zeta^{-1}$. The Gaussian binomial is palindromic of degree $s(r - s)$, so $[s]_{q^{-1}} = q^{-s(r-s)} [s]_q$, while $[r]_{q^{-1}} = (1 - q^{-r})/(1 - q^{-1}) = q^{1-r} [r]_q$. Hence

$$\overline{\rho_{r,s}} = \zeta^{r-1-s(r-s)} \rho_{r,s}.$$

A rational number equals its conjugate, and $\rho_{r,s} \neq 0$ may be cancelled, leaving $\zeta^{r-1-s(r-s)} = 1$. $\square$

Corollary 5.2. *For $2 \leq r < m$ the q -integer $[r]_\zeta$ is irrational. Hence $\rho_{r,0} = \rho_{r,r} = 1/[r]_\zeta$ is irrational for $2 \leq r < m$.*

Proof. Apply Proposition 5.1 with $s = 0$: rationality of $1/[r]_\zeta$, equivalently of $[r]_\zeta$, would force $m \mid r - 1$, impossible for $2 \leq r < m$. $\square$

The case $r = m - 1$ is completely explicit, and it is exactly where rationality does occur.

Proposition 5.3. *For $0 \leq s \leq m - 1$,*

$$\rho_{m-1,s} = (-1)^{s+1} \zeta^{1-s(s+1)/2},$$

an algebraic number of modulus 1. It is rational if and only if $\zeta^{1-s(s+1)/2} = \pm 1$, and its value is then ± 1 .

Proof. By (6), $[m-1]_q|_\zeta = (-1)^s \zeta^{-s(s+1)/2}$, while $[m-1]_\zeta = (1 - \zeta^{-1})/(1 - \zeta) = -\zeta^{-1}$. Divide. $\square$

Remark 5.4 (the irrationality hypothesis is false). An earlier version of this work assumed that $\rho_{r,s}$ is irrational for every $2 \leq r < m$ and every bad $s \in B(r)$. Proposition 5.3 refutes this. The smallest counterexample is $m = 9$, $r = 8$, $s = 4$, where $1 - s(s+1)/2 = -9 \equiv 0 \pmod{9}$ and hence $\rho_{8,4} = (-1)^5 = -1$. An exhaustive search — made feasible by Proposition 5.1, which confines the candidates to a congruence class — finds exactly 67 pairs with $m \leq 60$ and $s \in B(r)$ for which $\rho_{r,s}$ is rational. Every one has $r = m - 1$, and every value is ± 1 ; the first few are listed in Table 1. The consequences for the converse direction are dealt with in Section 9.

What survives is the following restricted statement, which is what Section 9 actually requires. It has been verified for all $m \leq 60$ and all admissible r, s , using Proposition 5.1 to prune the search.

m	r	bad s with $\rho_{r,s} \in \mathbb{Q}$	$\rho_{r,s}$
9	8	4	-1
10	9	3, 6	-1
14	13	5, 8	+1
18	17	4, 13	+1
18	17	7, 10	-1
20	19	6, 13	-1

Table 1: The first counterexamples to the irrationality hypothesis: bad residues at which $\rho_{r,s}$ is nevertheless rational. All of them have $r = m - 1$.

Conjecture 5.5. *Let ζ be a primitive m -th root of unity and $2 \leq r \leq m - 2$. Then $\rho_{r,s}$ is irrational for every bad residue $s \in B(r)$.*

Remark 5.6. A weaker statement suffices for Theorem 9.4, namely: *if $\rho_{r,s}$ is rational then $|\rho_{r,s}| = 1$.* Proposition 5.3 proves this for $r = m - 1$, and it holds in every case computed.

6 The criterion

Definition 6.1. Let $n \geq 3$ and $v \in (\mathbb{Z}/n\mathbb{Z})^\times$ of order $m > 1$, and put $f_e = \gcd(v^e - 1, n)$. We say that v satisfies the *criterion* if for every proper divisor e of m ,

- (i) $\varphi(f_e) \leq 2$, equivalently $f_e \in \{1, 2, 3, 4, 6\}$; and
- (ii) $\text{rad}(f_e) \mid m$.

We say that v *sieves* if

$$Y_{n,k}(\zeta_m) = \#\{k\text{-subset necklaces of } \mathbb{Z}/n\mathbb{Z} \text{ fixed by } v\} \quad \text{for every } k \text{ coprime to } n.$$

Since $f_e \mid f_{e'}$ whenever $e \mid e'$, the criterion need only be tested at the maximal proper divisors of m .

Lemma 6.2. *v satisfies the criterion if and only if $g_p := f_{m/p} \in \{1, 2, 3, 4, 6\}$ and $\text{rad}(g_p) \mid m$ for every prime $p \mid m$.*

Proof. Necessity is immediate. Conversely, every proper divisor e of m divides m/p for some prime $p \mid m$, so $f_e \mid g_p$; every divisor of an element of $\{1, 2, 3, 4, 6\}$ again lies in $\{1, 2, 3, 4, 6\}$, and $\text{rad}(f_e) \mid \text{rad}(g_p) \mid m$. $\square$

The two implications are in very different states, and we separate them.

Theorem 6.3 (sufficiency). *If v satisfies the criterion then v sieves.*

Theorem 6.3 is proved in Section 8. It applies to every multiplier, with no uniformity hypothesis. The converse is established in Section 9 for uniform multipliers with $f < m$, modulo Conjecture 5.5; in general it is verified but unproved.

Conjecture 6.4 (converse). *If v sieves then v satisfies the criterion.*

Since the cyclic sieving phenomenon of Definition 1.1 for the group $\langle u \rangle$ requires the identity at every power u^d , and u^d is itself a multiplier, Theorem 6.3 determines the full CSP by conjunction over the divisors of $\text{ord}_n(u)$.

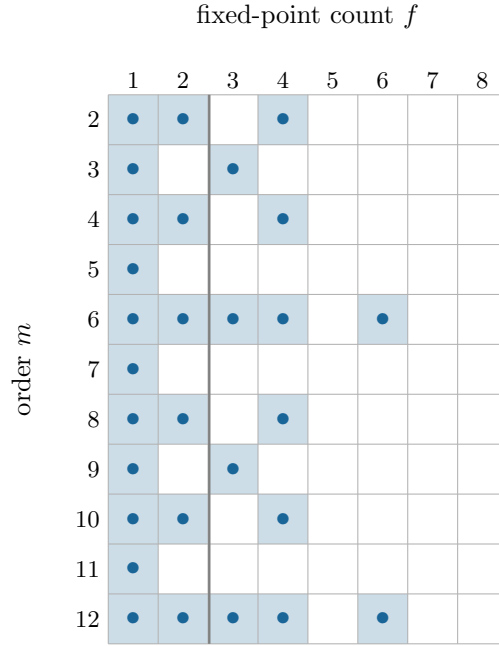


Figure 1: Uniform multipliers: sieving holds (●, shaded) or fails (blank), by the order m of the multiplier and its fixed-point count f . Columns $f = 5, 7, 8$ are empty for every m ; the pattern of a row depends only on which primes divide m . Columns $f = 1, 2$ are the previously known cases, columns $f = 3, 4, 6$ are new.

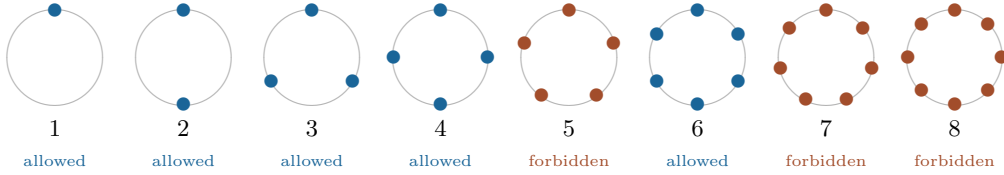


Figure 2: The permitted sizes of a multiplier's fixed set, $f \in \{1, 2, 3, 4, 6\}$ (Theorem 9.4) — identical to the permitted orders of rotational symmetry of a plane lattice, and for the same underlying reason: a quantity built from a root of unity is required to be rational.

Corollary 6.5 (uniform case). *If v is uniform then $f_e = f$ for every proper divisor e of m , and the criterion reads: $f \in \{1, 2, 3, 4, 6\}$ and $\text{rad}(f) \mid m$. In particular the verdict depends only on the pair (m, f) — not on n , and not on the number of orbits.*

Proof. If $e \mid m$ is proper and $v^e x = x$, the $\langle v \rangle$ -orbit of x has size dividing $e < m$, so by uniformity it has size 1. Hence $\text{Fix}(v^e) = \text{Fix}(v)$ and $f_e = f$. $\square$

The independence from n asserted in Corollary 6.5 is a striking feature of the data and was the first indication that a clean statement existed; see Figure 1.

7 What the criterion says

Write

$$P = \{ y \in \mathbb{Z}/n\mathbb{Z} : \text{the } \langle v \rangle\text{-orbit of } y \text{ has size } < m \} = \bigcup_{p|m \text{ prime}} \text{Fix}(v^{m/p}),$$

the second equality because a period smaller than m divides m/p for some prime $p \mid m$. Each $\text{Fix}(v^e)$ is the unique subgroup of the cyclic group $\mathbb{Z}/n\mathbb{Z}$ of order f_e , so P is a union of cyclic subgroups of orders $g_p = f_{m/p}$.

Lemma 7.1 (no clash). *Assume v satisfies the criterion. Then the integers g_p , for $p \mid m$ prime, are totally ordered by divisibility.*

Proof. By Lemma 6.2 every g_p lies in $\{1, 2, 3, 4, 6\}$, whose incomparable pairs are $\{2, 3\}$, $\{3, 4\}$ and $\{4, 6\}$. Two facts are used throughout: $g_p \mid n$; and if $2 \mid n$ then v is odd, so $2 \mid v - 1 \mid v^e - 1$ and hence $2 \mid f_e$ for every e .

$\{2, 3\}$. If $g_p = 2$ then $2 \mid n$, so every f_e is even, contradicting the oddness of $g_q = 3$.

$\{3, 4\}$. If $g_q = 4$ then $4 \mid n$, so again every f_e is even, contradicting $g_p = 3$.

$\{4, 6\}$. Suppose $g_p = 4$ and $g_q = 6$ with $p \neq q$. Then $4 \mid n$ and $6 \mid n$, so $12 \mid n$. Since $\gcd(v^{m/p} - 1, n) = 4$ and $3 \mid n$, we get $3 \nmid v^{m/p} - 1$; hence $\text{ord}_3(v) \neq 1$, so $\text{ord}_3(v) = 2$ and $2 \nmid m/p$. Since $\gcd(v^{m/q} - 1, n) = 6$ and $4 \mid n$, we get $4 \nmid v^{m/q} - 1$; hence $\text{ord}_4(v) = 2$ and $2 \nmid m/q$. But $3 \mid n$ forces $\text{ord}_3(v) \mid m$, so $2 \mid m$; then $2 \nmid m/p$ forces $p = 2$, and likewise $q = 2$, contradicting $p \neq q$. $\square$

Theorem 7.2 (structure). *Let $v \in (\mathbb{Z}/n\mathbb{Z})^\times$ have order $m > 1$ and put $L = \max_p g_p$. If v satisfies the criterion then P is precisely the cyclic subgroup $C_L \leq \mathbb{Z}/n\mathbb{Z}$ of order L , with*

$$L \in \{1, 2, 3, 4, 6\}, \quad \text{rad}(L) \mid m,$$

and v acts on C_L as $+1$ or -1 . Conversely, if P is a subgroup of order L with $\varphi(L) \leq 2$ and $\text{rad}(L) \mid m$, then v satisfies the criterion.

Proof. Assume the criterion. By Lemma 7.1 the subgroups $\text{Fix}(v^{m/p}) = C_{g_p}$ are nested, so their union P is the largest of them, namely C_L ; and $L = g_{p_0}$ lies in $\{1, 2, 3, 4, 6\}$ with $\text{rad}(L) \mid m$ by Lemma 6.2. The subgroup C_L is $\langle v \rangle$ -stable, and under the isomorphism $C_L \cong \mathbb{Z}/L\mathbb{Z}$ the multiplier acts as multiplication by $v \bmod L$, an element of $(\mathbb{Z}/L\mathbb{Z})^\times$. That group has order $\varphi(L) \leq 2$ and therefore equals $\{\pm 1\}$.

Conversely, if P is a subgroup of order L with $\varphi(L) \leq 2$ and $\text{rad}(L) \mid m$, then for every prime $p \mid m$ the group $\text{Fix}(v^{m/p})$ is a subgroup of P , so $g_p \mid L$. Every divisor of an element of $\{1, 2, 3, 4, 6\}$ again lies in that set, and $\text{rad}(g_p) \mid \text{rad}(L) \mid m$; apply Lemma 6.2. $\square$

Theorem 7.2 is a reformulation rather than a consequence: *the criterion says exactly that the points of period less than m form a subgroup of crystallographic order whose radical divides m . Since v then acts on that subgroup as ± 1 , its orbits have size 1 or 2, and only eight configurations are possible; all eight occur (Figure 3).*

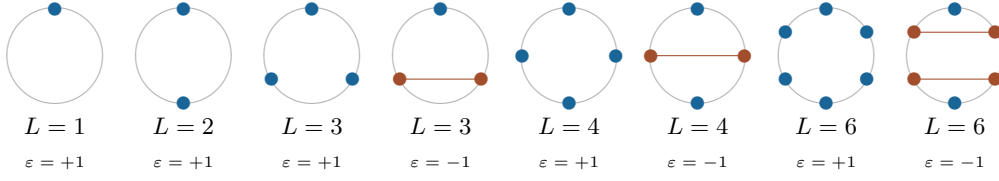


Figure 3: The eight admissible shapes of the non-generic part $P = C_L$ under the criterion (Theorem 7.2). Points fixed by the multiplier are drawn in azurite; the two-element orbits, which occur when v acts as -1 , are drawn in oxide and joined by their chord. Every criterion-satisfying multiplier with $n \leq 400$ realises one of these eight, and each of the eight is realised.

8 Proof of sufficiency

Throughout this section v satisfies the criterion, and L and C_L are as in Theorem 7.2. Write $f = f_1$, let n_m be the number of $\langle v \rangle$ -orbits of size m , and set

$$Q(z) = \prod_{|O| < m} (1 + z^{|O|}),$$

the product taken over the orbits contained in P . Then $n = L + mn_m$, and Corollary 3.3 reads

$$\# \text{Fix}(v) = \frac{1}{f} [z^k] N(z), \quad N(z) = Q(z) (1 + z^m)^{n_m}. \quad (7)$$

Identity (7), and the expansion (8) below, are formal consequences of Corollary 3.3 and the definition of Q ; they hold for every multiplier. Only the three lemmas that follow, and the proof of Theorem 6.3, invoke the criterion.

Lemma 8.1. *Q is palindromic of degree L , and $[z^1]Q = [z^{L-1}]Q = f$.*

Proof. Each factor $1 + z^{|O|}$ is palindromic, hence so is the product, and $\deg Q = \sum_{|O| < m} |O| = |P| = L$. The coefficient of z^1 counts the orbits of size 1, that is $|\text{Fix}(v)| = f$, and palindromicity gives $[z^{L-1}]Q = [z^1]Q$. $\square$

Lemma 8.2. *Let $s = k \bmod m$ for some k coprime to n . Then $\gcd(s, \text{rad}(L)) = 1$. If in addition $L \geq 2$ and $0 \leq s \leq L$, then $s \in \{1, L-1\}$.*

Proof. Let p be a prime dividing $\text{rad}(L)$. Then $p \mid L \mid n$ and $p \mid \text{rad}(L) \mid m$; if $p \mid s$ then $p \mid k$, contradicting $\gcd(n, k) = 1$. For the second claim, $s = 0$ and $s = L$ are divisible by every prime factor of L and so are excluded. It remains to see that every s with $2 \leq s \leq L-2$ shares a prime factor with L ; this range is empty for $L = 2, 3$, consists of $s = 2$ for $L = 4$, and of $s = 2, 3, 4$ for $L = 6$, and in each case the assertion holds. It is precisely here that $\varphi(L) \leq 2$ is used: for $L = 5$ the residue $s = 2$ would survive, and for $L \geq 7$ some prime below $L-1$ fails to divide L . $\square$

Lemma 8.3. *Put $r = n \bmod m$. Then $r = L \bmod m$, and exactly one of the following holds:*

- (a) $m > L$; then $r = L \geq 1$ and $m \nmid n$;
- (b) $m \leq L$; then $(L, m) \in \{(2, 2), (3, 3), (4, 2), (4, 4), (6, 6)\}$, $r = 0$ and $m \mid n$.

Proof. $n = L + mn_m$ gives $r = L \bmod m$. In case (a), $r = L$, and $L \geq 1$ because $0 \in P$. In case (b), $L \in \{1, 2, 3, 4, 6\}$ with $\text{rad}(L) \mid m$ and $2 \leq m \leq L$ leaves exactly the five listed pairs, and in each $m \mid L$, so $r = 0$. $\square$

Proof of Theorem 6.3. Fix k coprime to n and put $b = \lfloor k/m \rfloor$ and $s = k \bmod m$. By (7) it suffices to show $[z^k]N(z) = f Y_{n,k}(\zeta_m)$. Expanding (7) and substituting $i = b - j$ for the exponent of $(1 + z^m)^{n_m}$,

$$[z^k]N = \sum_{j \geq 0} \binom{n_m}{b-j} [z^{jm+s}]Q. \quad (8)$$

Case (a): $m \nmid n$. By Lemma 8.3, $r = L < m$, and $n_m = \lfloor n/m \rfloor =: a$ because $n = L + mn_m$ with $L < m$. Every term of (8) with $j \geq 1$ vanishes, since then $jm + s \geq m > L = \deg Q$; hence $[z^k]N = \binom{a}{b} [z^s]Q$. On the other side (4) gives $Y_{n,k}(\zeta_m) = \binom{a}{b} \rho_{L,s}$, so it suffices to prove $[z^s]Q = f \rho_{L,s}$. If $s > L$ then both sides vanish, because $\deg Q = L$ and $\lfloor \frac{L}{s} \rfloor_q = 0$. If $L \geq 2$ and $s \leq L$ then $s \in \{1, L-1\}$ by Lemma 8.2, whence $\rho_{L,s} = 1$ by Corollary 4.3, while $[z^s]Q = f$ by Lemma 8.1. If $L = 1$ then $f = 1$, $Q = 1 + z$, and $\rho_{1,s} = 1$ for $s \in \{0, 1\}$, so both sides equal 1.

Case (b): $m \mid n$. Here $r = 0$, $u := L/m \geq 1$ is an integer and $a := n/m = n_m + u$. By Vandermonde's identity $\binom{a-1}{b} = \sum_{j \geq 0} \binom{u-1}{j} \binom{n_m}{b-j}$, so comparing (8) with (5) it suffices to prove

$$[z^{jm+s}]Q = f \sigma_s \binom{u-1}{j} \quad \text{for every } j \geq 0. \quad (9)$$

For each of the five pairs $\text{rad}(L) = \text{rad}(m)$, so Lemma 8.2 says that the attainable residues are those $s \in \{0, \dots, m-1\}$ coprime to m , namely $s \in \{1, m-1\}$. Now $\sigma_1 = 1$, and by (6) together with $[m-1]_\zeta = -\zeta^{-1}$,

$$\sigma_{m-1} = (-1)^{m-1} \zeta^{1-(m-1)(m-2)/2} = 1 \quad \text{for } m \in \{2, 3, 4, 6\},$$

the four values of the exponent being 1, 0, -2, -9 against ζ of order 2, 3, 4, 6 respectively. Hence (9) reduces to $[z^{jm+s}]Q = f \binom{u-1}{j}$.

If $u = 1$, that is $(L, m) \in \{(2, 2), (3, 3), (4, 4), (6, 6)\}$, then $\binom{0}{j}$ vanishes unless $j = 0$. For $j = 0$ we need $[z^s]Q = f$ with $s \in \{1, m-1\} = \{1, L-1\}$, which is Lemma 8.1; for $j \geq 1$ we have $jm + s > m = L = \deg Q$ and both sides vanish.

If $(L, m) = (4, 2)$ then $u = 2$ and $s = 1$, and (9) reads $[z^{2j+1}]Q = f \binom{1}{j}$. For $j = 0$ this is $[z^1]Q = f$ and for $j = 1$ it is $[z^3]Q = [z^{L-1}]Q = f$, both by Lemma 8.1; for $j \geq 2$ we have $2j + 1 > 4 = \deg Q$ and both sides vanish. $\square$

Remark 8.4. The argument is short because Theorem 7.2 has converted the criterion into two properties of the single polynomial Q : it is palindromic of degree L with linear coefficient f , and the residues attainable subject to $\gcd(n, k) = 1$ are exactly $\{1, L-1\}$. The second is the crystallographic restriction. At no point does the proof need to know which of 1, 2, 3, 4, 6 the integer L is.

9 The crystallographic restriction

The plane crystallographic restriction says that a lattice in $\mathbb{R}^2$ admits rotational symmetry only of order 1, 2, 3, 4 or 6, because the trace $2 \cos(2\pi/f)$ of the rotation must be

a rational integer, i.e. $\varphi(f) \leq 2$. We now show that the same five numbers are forced here, by a different rationality constraint.

Lemma 9.1 (which residues occur). *Let $g = \gcd(m, n)$ and $0 \leq s < m$. There exists k with $k \equiv s \pmod{m}$ and $\gcd(k, n) = 1$ if and only if $\gcd(s, g) = 1$.*

Proof. If a prime p divides both s and g then $p \mid n$ and $p \mid k$ for every $k \equiv s \pmod{m}$, so no such k is coprime to n . Conversely assume $\gcd(s, g) = 1$ and let $p \mid n$ be prime. If $p \mid m$ then $p \mid g$, so $p \nmid s$ and hence $p \nmid k$ automatically for every $k \equiv s \pmod{m}$. If $p \nmid m$ then the progression $s + m\mathbb{Z}$ meets every residue class modulo p , so we may avoid 0; the finitely many such conditions are simultaneously satisfiable by the Chinese remainder theorem. $\square$

Lemma 9.2 (uniformity constrains primes). *Let v be uniform of order m , and let p be a prime with $p \mid n$ and $p \nmid f$. Then $\text{ord}_p(v) = m$; in particular $m \mid p - 1$ and $m \leq p - 1$.*

Proof. Let $H = (n/p)\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/p\mathbb{Z}$, a $\langle v \rangle$ -stable subgroup, and take $x = (n/p)u \in H$ with $p \nmid u$. Then $v^j x = x$ iff $n \mid (v^j - 1)(n/p)u$ iff $p \mid v^j - 1$, so the $\langle v \rangle$ -orbit of x has size $\text{ord}_p(v)$. If x were fixed then $p \mid v - 1$ and $p \mid n$, so $p \mid \gcd(v - 1, n) = f$, contrary to hypothesis. Hence x is not fixed, and uniformity forces its orbit size to be m , i.e. $\text{ord}_p(v) = m$. As $\text{ord}_p(v)$ divides $p - 1$, the claim follows. $\square$

Lemma 9.3 (arithmetic). *Let $f \geq 1$ be an integer such that every prime $q \leq f$ with $q \neq f - 1$ divides f . Then $f \in \{1, 2, 3, 4, 6\}$.*

Proof. The values $f = 1, 2, 3, 4, 6$ satisfy the condition: for $f = 6$ the primes ≤ 6 are 2, 3, 5 and $5 = f - 1$ is exempt, while $2 \mid 6$ and $3 \mid 6$; the others are immediate. For $f = 5$ the primes ≤ 5 are 2, 3, 5 and $f - 1 = 4$ is not prime, so $2 \mid 5$ would be required, which fails.

Now suppose $f \geq 7$. Every prime $q \leq f/2$ satisfies $q \neq f - 1$, because $f - 1 > f/2$ for $f > 2$; hence every prime $q \leq f/2$ divides f , and therefore

$$f \geq \prod_{p \leq f/2} p = e^{\theta(f/2)},$$

where θ is Chebyshev's function. Iterating from $f \geq 7$: primes $2, 3 \leq f/2$ give $6 \mid f$, so $f \geq 12$; then $2, 3, 5 \leq f/2$ give $30 \mid f$, so $f \geq 30$; then $2, 3, 5, 7, 11, 13 \leq f/2$ give $30030 \mid f$, so $f \geq 30030$. For $f \geq 30030$ the Chebyshev bound $\theta(x) > 0.84x$ (valid for $x \geq 101$) yields $f \geq e^{0.42f}$, which is false. Hence no $f \geq 7$ satisfies the condition. $\square$

Theorem 9.4 (necessity of the crystallographic restriction). *Let v be a uniform multiplier of order m with $f < m$, and assume Conjecture 5.5. If v sieves, then*

$$f \in \{1, 2, 3, 4, 6\} \quad \text{and} \quad \text{rad}(f) \mid m.$$

When $f = m - 1$ the appeal to Conjecture 5.5 may be dropped.

Proof. Uniformity gives $n = f + ma$, where a is the number of orbits of size m ; since $f < m$ we have $r := n \bmod m = f$, and moreover $P = \text{Fix}(v)$, $Q(z) = (1 + z)^f$ and $n_m = a$. Let $s = k \bmod m$ and $b = \lfloor k/m \rfloor$; then $b \leq \lfloor (n - 1)/m \rfloor = a$, so $\binom{a}{b} \geq 1$. Exactly as in Case (a) of the proof of Theorem 6.3 — an argument which uses only $\deg Q = f < m$, and not the criterion — identities (8) and (4) give

$$[z^k]N = \binom{a}{b} \binom{f}{s}, \quad f Y_{n,k}(\zeta_m) = f \binom{a}{b} \rho_{f,s},$$

so by (7) the sieving identity at k is equivalent to

$$\rho_{f,s} = \binom{f}{s} / f. \quad (10)$$

In particular $\rho_{f,s}$ must be *rational* for every attainable s . We show that no s in the bad set $B = B(f) = \{0, 2, 3, \dots, f-2, f\}$ is attainable; note that B is empty unless $f \geq 2$.

For $s \in \{0, f\}$ we have $\rho_{f,s} = 1/[f]_\zeta$, so (10) would force $[f]_\zeta = f \in \mathbb{Q}$, contradicting Corollary 5.2 since $2 \leq f < m$. This case is unconditional.

For $2 \leq s \leq f-2$ — a range non-empty only when $f \geq 4$ — the right-hand side of (10) obeys

$$\binom{f}{s} / f \geq \binom{f}{2} / f = \frac{f-1}{2} > 1.$$

If $f \leq m-2$, then $\rho_{f,s}$ is irrational by Conjecture 5.5 and (10) fails. If $f = m-1$, then $|\rho_{f,s}| = 1$ by Proposition 5.3, and (10) fails on absolute values. In either case s is not attainable.

Hence no k coprime to n has $k \bmod m \in B$. By Lemma 9.1 this says $\gcd(s, g) > 1$ for every $s \in B$, where $g = \gcd(m, n)$.

In particular, for every prime q with $2 \leq q \leq r = f$ and $q \neq f-1$ we have $q \in B$ and therefore $q \mid g$; so $q \mid m$ and $q \mid n$. Suppose such a q did not divide f . Then Lemma 9.2 applies to the prime q and gives $m \leq q-1$; but $q \mid m$ gives $q \leq m$, so $q \leq m \leq q-1$, a contradiction. Hence every prime $q \leq f$ with $q \neq f-1$ divides f , and Lemma 9.3 yields $f \in \{1, 2, 3, 4, 6\}$.

Finally we check $\text{rad}(f) \mid m$. Let q be any prime dividing f . Then $q \neq f-1$: otherwise q would divide both f and $f-1$, hence $q \mid 1$. So q falls under the case already treated, giving $q \mid g \mid m$. As this holds for every prime divisor of f , we conclude $\text{rad}(f) \mid m$. $\square$

Remark 9.5 (where the false hypothesis was felt). The step that would fail without Proposition 5.3 is the treatment of $2 \leq s \leq f-2$ when $f = m-1$. The smallest multiplier at which it is actually needed is

$$n = 99, \quad v = 19,$$

for which $m = 10$, $f = 9 = m-1$, v is uniform, and $s = 3$ is attainable — take $k = 13$. The earlier argument would have asserted $Y_{99,13}(\zeta_{10}) \notin \mathbb{Q}$; in truth $\rho_{9,3} = -1$ by Proposition 5.3, so that

$$Y_{99,13}(\zeta_{10}) = \binom{9}{1} \rho_{9,3} = -9,$$

a rational — indeed negative — integer, whereas the true number of fixed necklaces is $\binom{9}{1} \binom{9}{3} / 9 = 84$. Sieving does fail here, as it must, but on numerical rather than rationality grounds. We note that $99 > 96$, the largest n reached by the verification of Section 10 at the time the hypothesis was formulated; neither the proof nor the computation could have detected the gap.

Remark 9.6. Theorem 9.4 covers uniform multipliers with $f < m$; together with Theorem 6.3 it settles that case completely, modulo Conjecture 5.5. When $f \geq m$ — in particular when $m \mid n$, where Proposition 4.2 degenerates and Proposition 4.4 must be used instead — the same conclusion is observed in every computed case, but the argument above does not apply verbatim. Conjecture 6.4 for non-uniform multipliers remains open.

Remark 9.7 (why the coincidence is not one). Both the lattice statement and Theorem 9.4 amount to demanding that a quantity built from a root of unity be rational: for a lattice it is the trace of a rotation, here it is a Gaussian binomial divided by a q -integer. In each case the demand confines the root of unity to a field of degree at most two over $\mathbb{Q}$, and the surviving orders are $\{1, 2, 3, 4, 6\}$. A necklace contains no lattice; what the two situations share is only the circle.

10 Computational verification

All computations use exact integer arithmetic, together with exact arithmetic in $\mathbb{Z}[q]/(\Phi_m(q))$ for root-of-unity evaluations. No floating point is used anywhere.

Method. The polynomial side is computed from Proposition 4.1 as a product of cyclotomic polynomials reduced modulo Φ_m ; the fixed-point side from Corollary 3.3. Both were cross-validated against independent routes: the factorisation against direct polynomial division of $\begin{bmatrix} n \\ k \end{bmatrix}_q$ by $[n]_q$ for all $n < 30$, and the fixed-point formula both against the affine sum of Lemma 3.1 over all $c \in \mathbb{Z}/n\mathbb{Z}$ and against brute-force enumeration of necklaces. Rationality of an element of $\mathbb{Z}[q]/(\Phi_m)$ is decided by comparing it with a rational multiple of the reduced denominator, so no root of unity is ever approximated.

Scope. Each multiplier is tested against *every* admissible k ; the “instances” column counts multipliers for the criterion rows and individual checks otherwise.

Statement	range	instances	counterex.
Thm. 3.2, two colours, criterion <i>not</i> assumed	$n \leq 60$	29,780	0
Thm. 3.2, three colours, every content	$n \leq 28$	34,186	0
Thm. 7.2, structure of P	$n \leq 400$	19,820	0
Coefficient identities of §8	$n \leq 200$	1,048,609	0
Sieving identity, criterion satisfied (Thm. 6.3)	$n \leq 130$	2,527	0
Sieving identity, criterion violated (Conj. 6.4)	$n \leq 130$	2,497	0
Uniform (m, f) form (Cor. 6.5)	$n \leq 115$	2,682	0
Cor. 5.2, $[r]_\zeta \notin \mathbb{Q}$	$m \leq 60$	1,711	0
Prop. 5.1, congruence necessary	$m \leq 60$	34,278	0
Conj. 5.5, restricted irrationality	$m \leq 60$	34,278	0

Table 2: Exhaustive verification. The rows for Theorems 3.2 and 7.2 test statements proved above and so are consistency checks on the implementation; the row for Conjecture 6.4 is genuine evidence, as is the last row. Every counterexample to the discarded irrationality hypothesis found in the search of Remark 5.4 has $r = m - 1$, and there are 67 of them with $m \leq 60$.

11 Boundaries and open problems

Uniformity is not necessary. Of 280 non-uniform multipliers with $n \leq 60$, exactly 38 satisfy the sieving identity; all of them have orbit-size profile $\{1, 2, m\}$. Theorem 7.2 explains that profile: a non-uniform multiplier meeting the criterion must act on C_L as -1 , and the orbits of -1 on a cyclic group have size 1 or 2. This is also why Definition 6.1 ranges over all divisors $e \mid m$ rather than taking the (m, f) form: the

single-level condition on $f = f_1$ is necessary but not sufficient, failing in 106 of 603 multipliers with $n \leq 46$.

Three colours is strictly stronger. For contents with $\ell \geq 3$ colours the criterion as stated is no longer correct: among 213 multipliers with $n \leq 28$ there are 25 for which the two-colour identity holds but the three-colour identity fails — the smallest at $n = 28$. In every discrepancy the criterion over-predicts. Theorem 3.2 holds verbatim for every content, so the counting side of the identity is unaffected and the whole discrepancy lies on the q -analogue side: it is the choice of q -multinomial, not the enumeration of fixed necklaces, that fails to generalise. The additional obstruction is not understood.

Open problems.

1. Prove Conjecture 5.5, or the weaker modulus statement of Remark 5.6, which would render Theorem 9.4 unconditional.
2. Prove Conjecture 6.4 for non-uniform multipliers, and extend Theorem 9.4 to $f \geq m$.
3. Determine the correct criterion for contents with $\ell \geq 3$ colours, given that by Theorem 3.2 the obstruction is purely q -analytic.

A An unrelated side result: orbit towers terminate

A sequence $a = (a_j)_{j \geq 1}$ of non-negative integers is *realisable* if $L(a)_j = \frac{1}{j} \sum_{d|j} \mu(j/d) a_d$ is again a sequence of non-negative integers; then a_j counts the points of period j of a dynamical system and $L(a)_j$ counts its orbits of length j — its “primes”. One may ask whether the primes of a system can themselves be the points of a system, indefinitely.

Theorem A.1. *Let $a \neq 0$ be a sequence of non-negative integers. Then $L^j(a)$ fails to be a sequence of non-negative integers for some $j \geq 1$.*

Proof. Let $N = \min\{j : a_j \neq 0\}$. If $N \geq 2$ then $a_j = 0$ for $j < N$, and this persists: if $c_j = 0$ for all $j < N$ then $L(c)_j = 0$ for $j < N$, while $L(c)_N = c_N/N$ since every proper divisor of N contributes 0. Hence $L^j(a)_N = a_N/N^j$, which is a positive integer for all j only if $a_N = 0$, a contradiction.

If $N = 1$, put $\lambda = a_1 > 0$ and $x_j = L^j(a)_2$. Since L fixes the first term, $L^j(a)_1 = \lambda$ for all j , and $x_{j+1} = (x_j - \lambda)/2$. Solving, $x_j = -\lambda + (a_2 + \lambda)2^{-j} \rightarrow -\lambda < 0$, so $x_j < 0$ for large j . $\square$

Thus the tower of “primes of the primes of the primes” is always finite, with height at most $\log_N(a_N)$ when $N \geq 2$. Note the asymmetry: the opposite direction, $a \mapsto (\sum_{d|j} d a_d)_j$, may be iterated forever.

B Attribution ledger

The results above emerged from a broad computational search, most of whose output was already known. Recording that is part of the result.

- **New.** The criterion of Definition 6.1, its sufficiency (Theorem 6.3) and Theorem 9.4; the cases $f \in \{3, 4, 6\}$ are not covered by [2], and lie exactly in the regime that paper flags as having no general conjecture.
- **New.** Theorem 3.2, in particular the vanishing of $N_\alpha(A_c)$ for $c \notin (v-1)\mathbb{Z}/n\mathbb{Z}$ and the resulting f_1 -to-one correspondence, valid for every colour content.
- **New.** Theorem 7.2, which recasts the criterion as a statement about a single subgroup of crystallographic order.
- **Corrected here.** The irrationality hypothesis assumed in the first version of this work is false; see Remark 5.4. Proposition 5.1, Corollary 5.2 and Proposition 5.3 replace it, and Remark 9.5 records the first multiplier at which the difference is felt.
- **Rediscovered.** Sieving for prime n : derived and proved here via q -Lucas before being located as [2, Cor. 3.3]. The result is Stucky's.
- **Rediscovered.** That $([n]_q/[n-mk]_q) \begin{bmatrix} n-mk \\ k \end{bmatrix}_q$ cyclically sieves the k -subsets of a cycle with all cyclic gaps $\geq m+1$ (verified to $n=16$).
- **Rediscovered.** That the central binomial, Apéry, Franel and Domb numbers are periodic-point counts; their orbit counts are catalogued in the OEIS as the A060165 family.
- **Observed, uncatalogued.** Necklaces equal to their own Burrows–Wheeler transform. For $n=2, \dots, 18$ their numbers are

$$3, 4, 4, 4, 7, 6, 5, 9, 5, 6, 9, 8, 8, 11, 11, 7, 15,$$

so among the 4116 necklaces of length 16 exactly 11 are BWT-fixed. We found no database entry for this sequence and offer no characterisation.

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